Technical Neural Networks Assignment 1

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## Assignment 1

1. **A lot of simple, interconnected processing units (neurons).** Many simple units working in parallel, not one complex one.
2. **Modifiable, plastic connections (synapses, synaptic weights).** Connections can change strength – this is how learning happens.
3. **Mapping from an input vector to a scalar output value.** Several inputs combined into one output.
4. **Nonlinear transfer function.** Output isn't a linear function of the input, e.g. a threshold.

## Assignment 2

**Transformer** / self-attention mechanism – not mentioned in the lecture. I came across it while reading up on large language models and image generation models in my own time, and traced it back to the original paper.

Reference: Vaswani et al., *"Attention Is All You Need"*, NeurIPS 2017.

## Assignment 3

- **1940–1960, Beginning:** McCulloch-Pitts neuron and Hebbian learning proposed, leading up to Rosenblatt's Perceptron.
- **1960–1982, Quiet Years:** interest reduced after Minsky & Papert showed the Perceptron can't solve problems like XOR.
- **1980–1990, Renaissance:** Hopfield networks and backpropagation discovered and multi-layer networks became trainable.
- **1990–1999, Consolidation:** neural networks became one of the normal ML tools, used in practice.
- **since 2000, 2nd start-up:** deep learning takes off because of big data and GPUs.

## Assignment 4

we know that a neuron sums the weighted inputs and fires if the sum is more than a threshold $\Theta_m$:

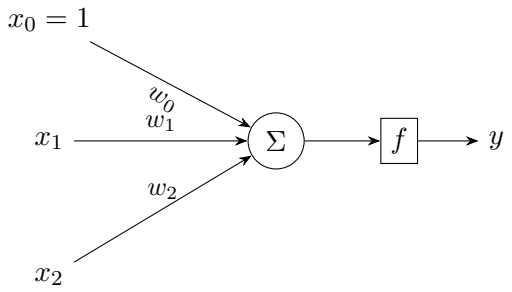
$$\text{net}_m^* = \sum_{n=1}^{N} w_{nm} x_n, \qquad y_m = 1 \text{ if } \text{net}_m^* > \Theta_m, \text{ else } 0.$$

Having a separate threshold for every neuron is very hard to handle and confusing. so we add one extra constant input $x_0 = 1$ with weight $w_{0m} = -\Theta_m$:

$$\text{net}_m = \sum_{n=0}^{N} w_{nm} x_n = \text{net}_m^* - \Theta_m, \qquad y_m = 1 \text{ if } \text{net}_m > 0.$$

we call the constant input $x_0 = 1$ BIAS. Also $w_{0m}$ is the BIAS-weight. It is just the negative threshold, added to the weights so it can be learned the same as other weights.

For 2 inputs the decision line is $w_1 x_1 + w_2 x_2 + w_0 = 0$– without $w_0$ this line would always go through the origin, so $w_0$ is what lets it shift away from it.




## Assignment 5

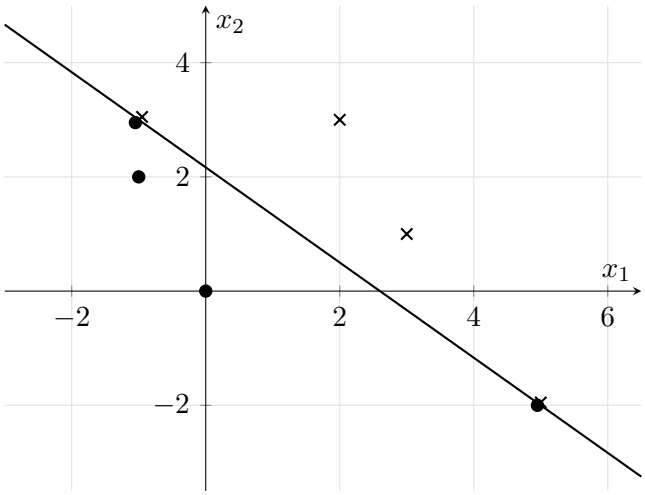
both the points: ${}^3X$ and ${}^4X$ sit right next to $(-1, 3)$ but on opposite sides of the line, so the line must pass exactly from $(-1, 3)$. Same thing is for $(5, -2)$ with ${}^7X$,${}^8X$. So $g$ is the line through $(-1, 3)$ and $(5, -2)$:

$$m = \frac{-2-3}{5-(-1)} = -\frac{5}{6}, \qquad x_2 - 3 = -\tfrac{5}{6}(x_1 + 1) \;\Rightarrow\; 5x_1 + 6x_2 - 13 = 0.$$

if we test it on ${}^1X$ -> $(0, 0; +1)$: $5(0) + 6(0) - 13 = -13 < 0$, but ${}^1X$ should be $+1$, so we should change the sign:

$$w_1 = -5, \quad w_2 = -6, \quad w_b = 13, \qquad g: \; -5x_1 - 6x_2 + 13 = 0.$$

| | $(x_1, x_2)$ | $-5x_1 - 6x_2 + 13$ | $\hat{y}$ |
|---|---|---|---|
| ${}^1X$ | $(0, 0)$ | $13$ | $+1$ |
| ${}^2X$ | $(2, 3)$ | $-15$ | $-1$ |
| ${}^3X$ | $(-1-\varepsilon, 3-\varepsilon)$ | $+11\varepsilon$ | $+1$ |
| ${}^4X$ | $(-1+\varepsilon, 3+\varepsilon)$ | $-11\varepsilon$ | $-1$ |
| ${}^5X$ | $(-1, 2)$ | $6$ | $+1$ |
| ${}^6X$ | $(3, 1)$ | $-8$ | $-1$ |
| ${}^7X$ | $(5, -2+\varepsilon)$ | $-6\varepsilon$ | $-1$ |
| ${}^8X$ | $(5-\varepsilon, -2)$ | $+5\varepsilon$ | $+1$ |




## Assignment 6

**AI use:** I used Claude to get help for this question. I gave it the assignment text directly and asked it to find weights for single threshold neurons solving AND, NAND, NOR and to check them against the truth tables. The weights it gave worked on the first try, so I didn't need to ask it for changes.

Single threshold neuron, $y = 1$ if $w_1 x_1 + w_2 x_2 > \Theta$, else 0:

| | $w_1$ | $w_2$ | $\Theta$ |
|---|---|---|---|
| AND | 1 | 1 | 1.5 |
| NAND | −1 | −1 | −1.5 |
| NOR | −1 | −1 | −0.5 |

| $x_1$ | $x_2$ | AND | | NAND | | NOR | |
|---|---|---|---|---|---|---|---|
| | | net | $y$ | net | $y$ | net | $y$ |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 | −1 | 1 | −1 | 0 |
| 1 | 0 | 1 | 0 | −1 | 1 | −1 | 0 |
| 1 | 1 | 2 | 1 | −2 | 0 | −2 | 0 |

All three match the truth tables.

